Technische Hochschule Ingolstadt

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In this Lecture we will cover:

Newton's Laws

- $\sum \vec{M}_i = \frac{d}{dt}\vec{H}$
- $\sum \vec{F}_i = m\frac{d}{dt}\vec{v}$

Rotating Frames of Reference

- Equations of Motion in Body-Fixed Frame
- Often Confusing

Review: Coordinate Rotations

Positive Directions

If in doubt, use the right-hand rules.

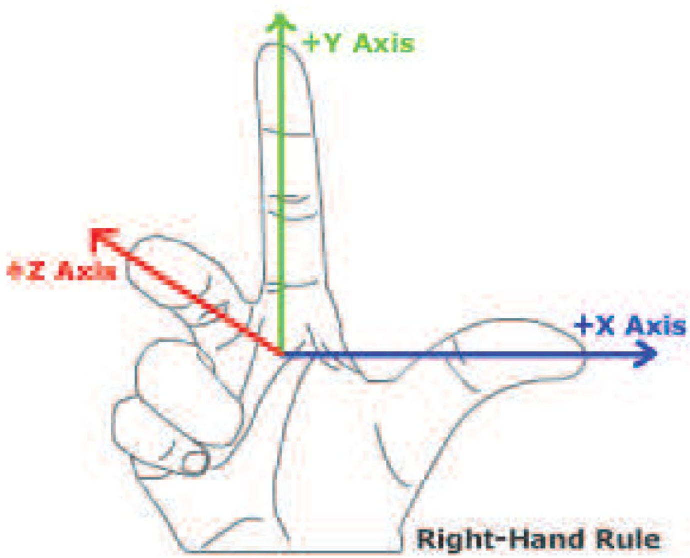


Figure: Positive Directions

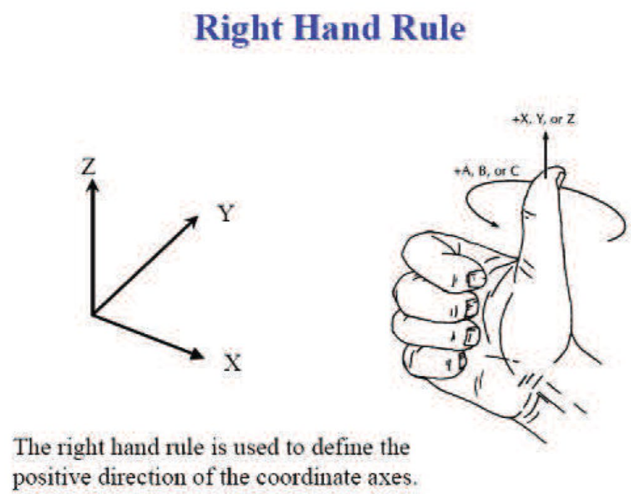
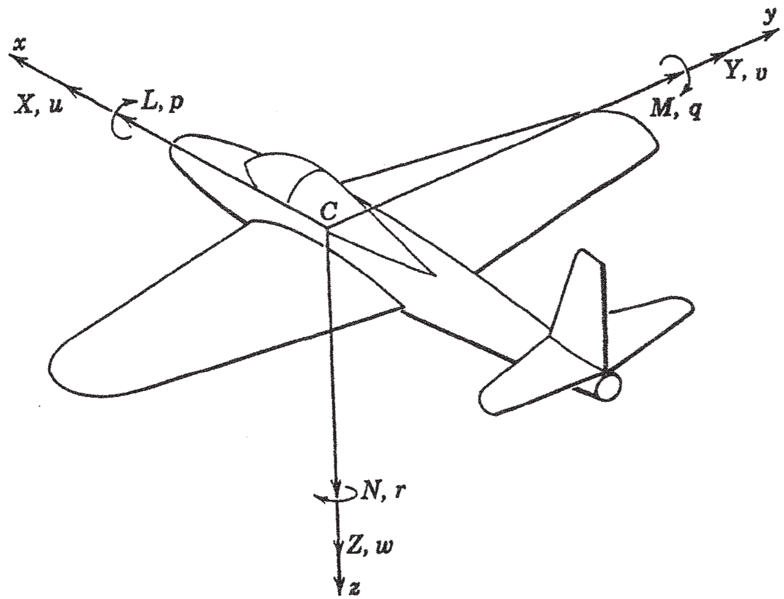


Figure: Positive Rotations

Review: Coordinate Rotations

Roll-Pitch-Yaw



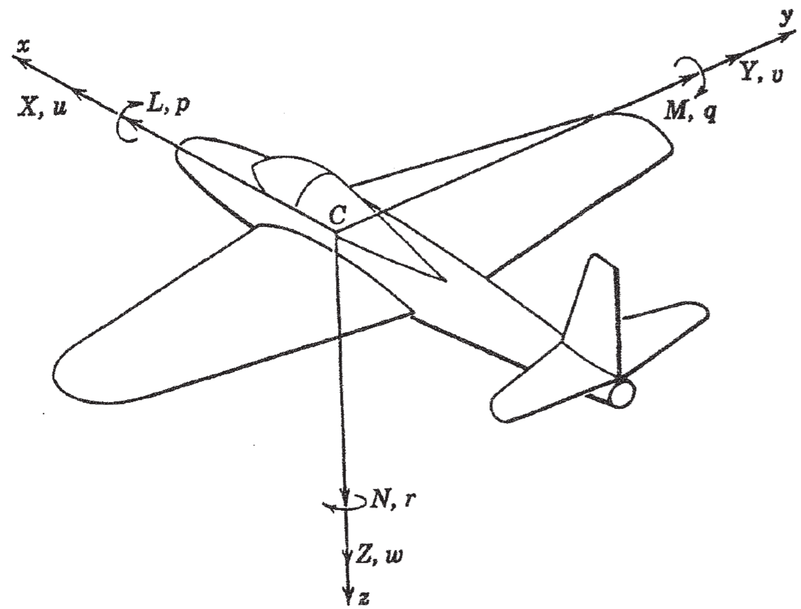
There are 3 basic rotations an aircraft can make:

- Roll = Rotation about x-axis
- Pitch = Rotation about y-axis
- Yaw = Rotation about z-axis
- Each rotation is a one-dimensional transformation.

Any two coordinate systems can be related by a sequence of 3 rotations.

Review: Forces and Moments

Forces

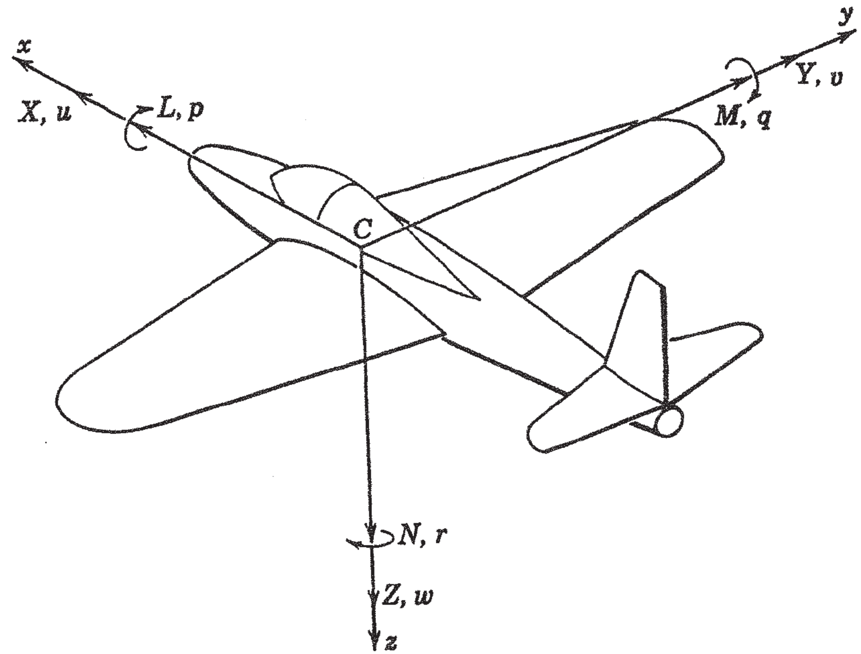


These forces and moments have standard labels. The Forces are:

X	Axial Force	Net Force in the positive x-direction
Y	Side Force	Net Force in the positive y-direction
Z	Normal Force	Net Force in the positive z-direction

Review: Forces and Moments

Moments



The Moments are called, intuitively:

L	Rolling Moment	Net Moment in the positive p-direction
M	Pitching Moment	Net Moment in the positive q-direction
N	Yawing Moment	Net Moment in the positive r-direction

6DOF: Newton's Laws

Forces

Newton's Second Law tells us that for a particle $F = ma$. In vector form:

$$\vec{F} = \sum_i \vec{F}_i = m \frac{d}{dt} \vec{V}$$

That is, if $\vec{F} = [F_x \ F_y \ F_z]$ and $\vec{V} = [u \ v \ w]$, then

$$F_x = m \frac{du}{dt} \qquad F_x = m \frac{dv}{dt} \qquad F_z = m \frac{dw}{dt}$$

Definition 1.

$\vec{L} = m\vec{V}$ is referred to as **Linear Momentum**.

Newton's Second Law is only valid if $\vec{F}$ and $\vec{V}$ are defined in an *Inertial* coordinate system.

Definition 2.

A coordinate system is **Inertial** if it is not accelerating or rotating.

Newton's Laws

Moments

Using Calculus, this concept can be extended to rigid bodies by integration over all particles.

$$\vec{M} = \sum_i \vec{M}_i = \frac{d}{dt}\vec{H}$$

Definition 3.

Where $\vec{H} = \int (\vec{r}_c \times \vec{v}_c) dm$ is the **angular momentum**.

Angular momentum of a rigid body can be found as

$$\vec{H} = I\vec{\omega}_I$$

where $\vec{\omega}_I = [p,\ q,\ r]^T$ is the angular rotation vector of the body about the center of mass.

- p is rotation about the x-axis.
- q is rotation about the y-axis.
- r is rotation about the z-axis.
- ω_I is defined in an *Inertial* Frame.

The matrix I is the *Moment of Inertia Matrix*.

Newton's Laws

Moment of Inertia

The moment of inertia matrix is defined as

$$I = \begin{bmatrix} I_{xx} & -I_{xy} & -I_{xz} \\ -I_{yx} & I_{yy} & -I_{yz} \\ -I_{zx} & -I_{zy} & I_{zz} \end{bmatrix}$$

$$I_{xy} = I_{yx} = \int\int\int xydm \qquad I_{xx} = \int\int\int \left(y^2 + z^2\right) dm$$

$$I_{xz} = I_{zx} = \int\int\int xzdm \qquad I_{yy} = \int\int\int \left(x^2 + z^2\right) dm$$

$$I_{yz} = I_{zy} = \int\int\int yzdm \qquad I_{zz} = \int\int\int \left(x^2 + y^2\right) dm$$

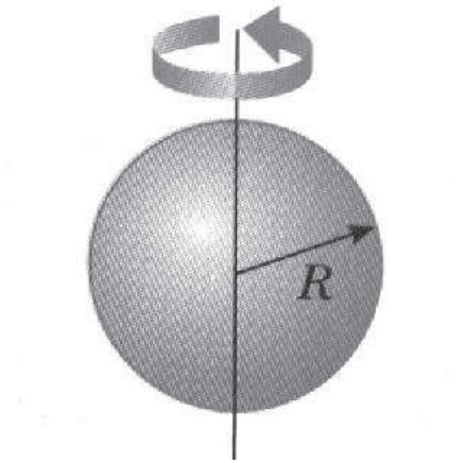
So

$$\begin{bmatrix} H_x \\ H_y \\ H_z \end{bmatrix} = \begin{bmatrix} I_{xx} & -I_{xy} & -I_{xz} \\ -I_{yx} & I_{yy} & -I_{yz} \\ -I_{zx} & -I_{zy} & I_{zz} \end{bmatrix} \begin{bmatrix} p_I \\ q_I \\ r_I \end{bmatrix}$$

where p_I, q_I and r_I are the rotation vectors as expressed in the inertial frame corresponding to x-y-z.

Moment of Inertia

Examples:



Homogeneous Sphere

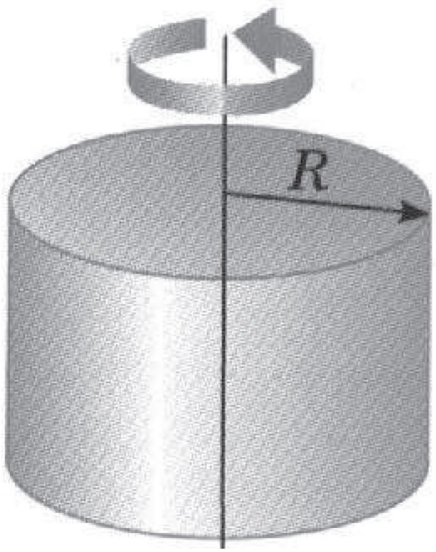
$$I_{sphere} = \frac{2}{5} m r^2 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Ring

$$I_{ring} = m r^2 \begin{bmatrix} \frac{1}{2} & 0 & 0 \\ 0 & \frac{1}{2} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Moment of Inertia

Examples:



Homogeneous Disk

$$I_{disk} = \frac{1}{4}mr^2 \begin{bmatrix} 1+\frac{1}{3}\frac{h}{r^2} & 0 & 0 \\ 0 & 1+\frac{1}{3}\frac{h}{r^2} & 0 \\ 0 & 0 & \frac{1}{2} \end{bmatrix}$$

F/A-18

$$I = \begin{bmatrix} 23 & 0 & 2.97 \\ 0 & 15.13 & 0 \\ 2.97 & 0 & 16.99 \end{bmatrix} kslug - ft^2$$

Problem:

The Body-Fixed Frame

The moment of inertia matrix, I, is fixed in the body-fixed frame. However, Newton's law only applies for an inertial frame:

$$\vec{M} = \sum_i \vec{M}_i = \frac{d}{dt}\vec{H}$$

If the body-fixed frame is rotating with rotation vector $\vec{\omega}$, then for any vector, $\vec{a}$, $\frac{d}{dt}\vec{a}$ in the inertial frame is

$$\left.\frac{d\vec{a}}{dt}\right|_I = \left.\frac{d\vec{a}}{dt}\right|_B + \vec{\omega} \times \vec{a}$$

Specifically, for Newton's Second Law

$$\vec{F} = m\left.\frac{d\vec{V}}{dt}\right|_B + m\vec{\omega} \times \vec{V}$$

and

$$\vec{M} = \left.\frac{d\vec{H}}{dt}\right|_B + \vec{\omega} \times \vec{H}$$

Equations of Motion

Thus we have

$$\begin{bmatrix} F_x \\ F_y \\ F_z \end{bmatrix} = m \begin{bmatrix} \dot{u} \\ \dot{v} \\ \dot{w} \end{bmatrix} + m \det \begin{bmatrix} \hat{x} & \hat{y} & \hat{z} \\ p & q & r \\ u & v & w \end{bmatrix} = m \begin{bmatrix} \dot{u} + qw - rv \\ \dot{v} + ru - pw \\ \dot{w} + pv - qu \end{bmatrix}$$

and

$$\begin{bmatrix} L \\ M \\ N \end{bmatrix} = \begin{bmatrix} I_{xx} & -I_{xy} & -I_{xz} \\ -I_{yx} & I_{yy} & -I_{yz} \\ -I_{zx} & -I_{zy} & I_{zz} \end{bmatrix} \begin{bmatrix} \dot{p} \\ \dot{q} \\ \dot{r} \end{bmatrix} + \vec{\omega} \times \begin{bmatrix} I_{xx} & -I_{xy} & -I_{xz} \\ -I_{yx} & I_{yy} & -I_{yz} \\ -I_{zx} & -I_{zy} & I_{zz} \end{bmatrix} \begin{bmatrix} p \\ q \\ r \end{bmatrix}$$

$$= \begin{bmatrix} I_{xx}\dot{p} - I_{xy}\dot{q} - I_{xz}\dot{r} \\ -I_{xy}\dot{p} + I_{yy}\dot{q} - I_{yz}\dot{r} \\ -I_{xz}\dot{p} - I_{yz}\dot{q} + I_{zz}\dot{r} \end{bmatrix} + \vec{\omega} \times \begin{bmatrix} pI_{xx} - qI_{xy} - rI_{xz} \\ -pI_{xy} + qI_{yy} - rI_{yz} \\ -pI_{xz} - qI_{yz} + rI_{zz} \end{bmatrix}$$

$$= \begin{bmatrix} I_{xx}\dot{p} - I_{xy}\dot{q} - I_{xz}\dot{r} + q(-pI_{xz} - qI_{yz} + rI_{zz}) - r(-pI_{xy} + qI_{yy} - rI_{yz}) \\ -I_{xy}\dot{p} + I_{yy}\dot{q} - I_{yz}\dot{r} - p(-pI_{xz} - qI_{yz} + rI_{zz}) + r(pI_{xx} - qI_{xy} - rI_{xz}) \\ -I_{xz}\dot{p} - I_{yz}\dot{q} + I_{zz}\dot{r} + p(-pI_{xy} + qI_{yy} - rI_{yz}) - q(pI_{xx} - qI_{xy} - rI_{xz}) \end{bmatrix}$$

Which is too much for any mortal. For aircraft, we have **symmetry** about the x-z plane. Thus $I_{yz} = I_{xy} = 0$. Spacecraft?

Equations of Motion

Reduced Equations

With $I_{xy} = I_{yz} = 0$, we have, in summary:

$$\begin{bmatrix} F_x \\ F_y \\ F_z \end{bmatrix} = m \begin{bmatrix} \dot{u} + qw - rv \\ \dot{v} + ru - pw \\ \dot{w} + pv - qu \end{bmatrix}$$

and

$$\begin{bmatrix} L \\ M \\ N \end{bmatrix} = \begin{bmatrix} I_{xx}\dot{p} - I_{xz}\dot{r} - qpI_{xz} + qrI_{zz} - rqI_{yy} \\ I_{yy}\dot{q} + p^2 I_{xz} - prI_{zz} + rpI_{xx} - r^2 I_{xz} \\ -I_{xz}\dot{p} + I_{zz}\dot{r} + pqI_{yy} - qpI_{xx} + qrI_{xz} \end{bmatrix}$$

Right now,

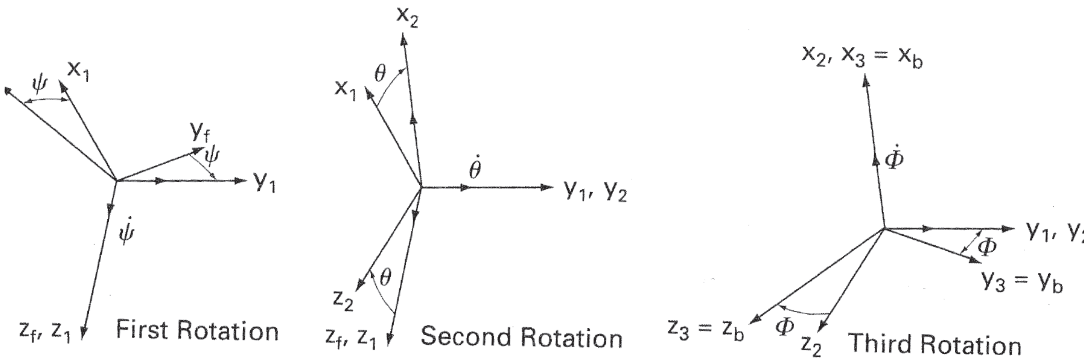
- Translational variables (u,v,w) depend on rotational variables (p,q,r).
- Rotational variables (p,q,r) do not depend on translational variables (u,v,w).
  - For aircraft, however, Moment forces (L,M,N) depend on rotational and translational variables.

Euler Angles

Issue: Equations of motion are expressed in the Body-Fixed frame.

Question: How do determine rotation and velocity in the inertial frame. For intercept, obstacle avoidance, etc.

Approach: From Lecture 4, any two coordinate systems can be related through a sequence of three rotations. Recall these transformations are:



Roll Rotation (ϕ) :

$R_1(\phi)$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\phi & -\sin\phi \\ 0 & \sin\phi & \cos\phi \end{bmatrix}$$

Pitch Rotation (θ):

$R_2(\theta)$

$$= \begin{bmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{bmatrix}$$

Yaw Rotation (ψ):

$R_3(\psi)$

$$= \begin{bmatrix} \cos\psi & -\sin\psi & 0 \\ \sin\psi & \cos\psi & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Euler Angles

Definition 4.

The term **Euler Angles** refers to the angles of rotation (ψ, θ, ϕ) needed to go from one coordinate system to another using the specific sequence of rotations **Yaw-Pitch-Roll**:

$$\vec{V}_{BF} = R_1(\phi)R_2(\theta)R_3(\psi)\vec{V}_I.$$

NOTE BENE: Euler angles are often defined differently (e.g. 3-1-3). We use the book notation.
The composite rotation matrix can be written

$$R_1(\phi)R_2(\theta)R_3(\psi) =$$
$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\phi & -\sin\phi \\ 0 & \sin\phi & \cos\phi \end{bmatrix} \begin{bmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{bmatrix} \begin{bmatrix} \cos\psi & -\sin\psi & 0 \\ \sin\psi & \cos\psi & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

This moves a vector

Inertial Frame $\Rightarrow$ Body-Fixed Frame

Euler Angles

Thus to find the inertial velocity vector, we must rotate **FROM** the body-fixed coordinates to the inertial frame:

$$\begin{bmatrix} \frac{dx}{dt} \\ \frac{dy}{dt} \\ \frac{dz}{dt} \end{bmatrix} = R_3(-\psi)R_2(-\theta)R_1(-\phi) \begin{bmatrix} u \\ v \\ w \end{bmatrix}$$

Euler Angles

The rate of rotation of the Euler Angles can be found by rotating the rotation vector into the inertial frame

$$\begin{bmatrix} p \\ q \\ r \end{bmatrix} = \begin{bmatrix} 1 & 0 & -\sin\theta \\ 0 & \cos\phi & \cos\theta\sin\phi \\ 0 & -\sin\theta & \cos\theta\cos\phi \end{bmatrix} \begin{bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix}$$

This transformation can also be reversed as

$$\begin{bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix} = \begin{bmatrix} 1 & \sin\phi\tan\theta & \cos\phi\tan\theta \\ 0 & \cos\phi & -\sin\phi \\ 0 & \sin\phi\sec\theta & \cos\phi\sec\theta \end{bmatrix} \begin{bmatrix} p \\ q \\ r \end{bmatrix}$$

Summary

$$X - mgS_\theta = m(\dot{u} + qw - rv)$$
$$Y + mgC_\theta S_\Phi = m(\dot{v} + ru - pw)$$
$$Z + mgC_\theta C_\Phi = m(\dot{w} + pv - qu)$$

Force equations

$$L = I_x\dot{p} - I_{xz}\dot{r} + qr(I_z - I_y) - I_{xz}pq$$
$$M = I_y\dot{q} + rp(I_x - I_z) + I_{xz}(p^2 - r^2)$$
$$N = -I_{xz}\dot{p} + I_z\dot{r} + pq(I_y - I_x) + I_{xz}qr$$

Moment equations

$$p = \dot{\Phi} - \dot{\psi}S_\theta$$
$$q = \dot{\theta}C_\Phi + \dot{\psi}C_\theta S_\Phi$$
$$r = \dot{\psi}C_\theta C_\Phi - \dot{\theta}S_\Phi$$

Body angular velocities in terms of Euler angles and Euler rates

$$\dot{\theta} = qC_\Phi - rS_\Phi$$
$$\dot{\Phi} = p + qS_\Phi T_\theta + rC_\Phi T_\theta$$
$$\dot{\psi} = (qS_\Phi + rC_\Phi)\sec\theta$$

Euler rates in terms of Euler angles and body angular velocities

Velocity of aircraft in the fixed frame in terms of Euler angles and body velocity components

$$\begin{bmatrix} \frac{dx}{dt} \\ \frac{dy}{dt} \\ \frac{dz}{dt} \end{bmatrix} = \begin{bmatrix} C_\theta C_\psi & S_\Phi S_\theta C_\psi - C_\Phi S_\psi & C_\Phi S_\theta C_\psi + S_\Phi S_\psi \\ C_\theta S_\psi & S_\Phi S_\theta S_\psi + C_\Phi C_\psi & C_\Phi S_\theta S_\psi - S_\Phi C_\psi \\ -S_\theta & S_\Phi C_\theta & C_\Phi C_\theta \end{bmatrix} \begin{bmatrix} u \\ v \\ w \end{bmatrix}$$